

# Experiment - 06

Student Name: Shreyansh Gupta  
 Branch: CSE - AIML  
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 Subject Name: ADBMS

UID: 24BAI70658  
 Section/Group: 24AIT\_KRG-1\_B  
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## (A)

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

Difference = Apples Sold - Oranges Sold

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

**Sales**

| sale_date  | fruit   | sold_num |
|------------|---------|----------|
| 2020-05-01 | apples  | 10       |
| 2020-05-01 | oranges | 8        |
| 2020-05-02 | apples  | 15       |
| 2020-05-02 | oranges | 15       |
| 2020-05-03 | apples  | 20       |
| 2020-05-03 | oranges | 0        |
| 2020-05-04 | apples  | 15       |
| 2020-05-04 | oranges | 16       |

**Output**

| sale_date  | diff |
|------------|------|
| 2020-05-01 | 2    |
| 2020-05-02 | 0    |
| 2020-05-03 | 20   |
| 2020-05-04 | -1   |

## Solution:

```
SELECT SALE_DATE, SUM(CASE WHEN FRUIT='apples' THEN SOLD_NUM
ELSE -SOLD_NUM END) AS DIFF
FROM SALES
GROUP BY SALE_DATE
```

ORDER BY SALE\_DATE ASC

**(B)**

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

id represents the current node.

p\_id represents the parent node.

Classify every node into one of the following categories:

Root: A node that does not have any parent.

Leaf: A node that has a parent but does not have any children.

Inner: A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

**Tree**

| id | p_id |
|----|------|
| 1  | NULL |
| 2  | 1    |
| 3  | 1    |
| 4  | 2    |
| 5  | 2    |

**Expected Output**

| id | Type  |
|----|-------|
| 1  | Root  |
| 2  | Inner |
| 3  | Leaf  |
| 4  | Leaf  |
| 5  | Leaf  |

**Solution:**

```

SELECT ID,
       CASE WHEN P_ID IS NULL THEN 'ROOT'
            WHEN ID NOT IN (SELECT P_ID FROM TREE WHERE P_ID IS NOT
NULL) THEN 'LEAF'
            ELSE 'INNER' END AS "TYPE"
FROM TREE

```